Dongho Lee

Center for Quantum Structures in Modules and Spaces,
Department of Mathematical Sciences, Seoul National University
Seoul, Republic of Korea
Homepage : https://dhlee-math.github.io
Mail : dhlee337@proton.me or tangled@snu.ac.kr

Research Interests

Keywords: Symplectic geometry; Floer theory; Hamiltonian dynamics; celestial mechanics.

My research focuses on Hamiltonian dynamics through the framework of symplectic geometry. More specifically, I study bifurcation phenomena of periodic Hamiltonian orbits using tools from Floer theory, with the goal of understanding how such local dynamical changes are reflected in symplectic and Floer-theoretic invariants. I am also interested in applications to celestial mechanics, especially restricted three-body problems and related Hamiltonian systems.

Employment

Postdoctoral Researcher May 2025 – Present
Center for Quantum Structures in Modules and Spaces (QSMS), Seoul National University

Postdoctoral Researcher Mar 2025 – Apr 2025
Research Institute of Mathematics, Seoul National University
Postdoctoral supervisor: Prof. Cheol-hyun Cho

Education

Ph.D. in Mathematics Mar 2018 – Feb 2025
Department of Mathematical Sciences, Seoul National University
Advisor: Prof. Otto van Koert

B.S. in Mathematics; B.A. in Economics; Minor in Physics Mar 2013 – Feb 2018
College of Liberal Studies, Seoul National University
Graduated with Honors

Publications and Preprints

1. **Conley-Zehnder Indices of Spatial Rotating Kepler Problem**.
   arXiv:2506.14325; *Journal of Topology and Analysis*, published online, 2026.

2. **Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces**.
   With Sunghae Cho; arXiv:2401.02667; *Archiv der Mathematik*, 123, 291–307, 2024.

Works in Progress

1. **Mutation of Local Floer Chain Complex across Hamiltonian Bifurcations**.
   With Hong-kwon Jo. Manuscript in preparation; expected Summer 2026.

2. **Fiberwise Convexity of the Circular Restricted Three-Body Problem**.
   With Sunghae Cho and Beomjun Sohn. Manuscript in preparation; expected Summer 2026.

Thesis

1. **Global Hypersurfaces of Section and the Spatial Kepler Problem**.
   Ph.D. thesis, Seoul National University, 2025.
   Advisor: Prof. Otto van Koert.

Invited Talks

1. **Local Floer Chain Complex across Hamiltonian Bifurcation**.
   Geometry Space Surrey, University of Surrey, Guildford, UK — 10 June 2026
2. **Conley-Zehnder Indices and Bifurcations of the Spatial Rotating Kepler Problem**.
   Oyster Dynamics Seminar, Dalian University of Technology, Dalian, China — 19 May 2026
3. **Bifurcation of Periodic Orbits in Hamiltonian Systems**.
   G-Lamp Mathematics Seminar, Soongsil University, Seoul, Korea — 22 Jan 2026
4. **Generic Bifurcation of Hamiltonian Orbits**.
   Symplectic Learning Seminar, Seoul National University, Seoul, Korea — 12 Dec 2025
5. **Hamiltonian Dynamics and Three-Body Problem**.
   Rookie's Pitch, Seoul National University, Seoul, Korea — 14 Oct 2025
6. **Three-Body Problem and Related Problems**.
   QSMS 2025 Summer Workshop, Goseong, Korea — 21 Aug 2025
7. **Periodic Orbits of Rotating Kepler Problem**.
   Billiards and Quantitative Symplectic Geometry, Heidelberg, Germany — 16 Jul 2025
8. **Closed Orbits of the Spatial Rotating Kepler Problem**.
   2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea — 25 Apr 2025

Poster Presentation

1. **Periodic Orbits and Symplectic Invariants in the Rotating Kepler Problem**.
   Celestial Mechanics & N-body Dynamics, Tokyo, Japan — 16 Nov 2025

Selected Conferences and Workshops

Geometric Structures and Differential Equations – Symmetry, Singularity and Applications
RIMS, Kyoto University, Kyoto, Japan — 13–17 Jul 2026

International Conference on Celestial Mechanics
SUSTech, Shenzhen, China — 20–24 Apr 2026

New Developments in Symplectic Geometry
Seoul National University, Seoul, Korea — 25–28 Nov 2025

East Asian Symplectic Conference 2025, Hokkaido University, Japan — 25–29 Sep 2025

2025 Symplectic Retreat, Tongyeong, Korea — 23–26 Jan 2025

From Hamiltonian Dynamics to Symplectic Topology and Beyond
Institut Henri Poincaré, Paris, France — 3–7 Jun 2024

Topology of 4-Manifolds and Related Topics Jeju, Korea — 21–27 Jan 2024

East Asian Symplectic Conference 2023, Jeju, Korea — 29 Oct–4 Nov 2023

Workshop on Symplectic Dynamics
Instituto Superior Técnico, Lisbon, Portugal — 27 Jun–1 Jul 2022

Fukaya 60: Geometry and Everything, Kyoto University, Japan — 17–22 Feb 2019

Teaching Experience

Teaching and Course Assistant, Seoul National University Mar 2018 – Feb 2025

- **Major and graduate courses:** Algebraic Topology (graduate), Introduction to Topology, Introduction to Differential Geometry, Selected Topics Seminar.
- **General mathematics courses:** Engineering Mathematics, Calculus, Calculus Practice, Calculus for Life Science, Fundamentals and Applications of Mathematics.
- **Head Teaching Assistant** for Introductory Mathematics Courses Spring 2020

Awards and Honors

Merit-based Scholarship
Department of Mathematical Sciences, Seoul National University 2018

National Scholarship, Korea Student Aid Foundation 2015

Merit-based Scholarship, College of Liberal Studies, Seoul National University 2013

Affiliations and Activities

Dynamical Systems: Limits and Recurrence, organizer 2026 – Present

Korean Mathematical Society, member 2025 – Present

QSMS-BK21 Symplectic Seminar, founding member 2021 – Present

References

Prof. Otto van Koert
Department of Mathematical Sciences
Seoul National University
`okoert@snu.ac.kr`
Ph.D. advisor

Prof. Cheol-hyun Cho
Department of Mathematics
POSTECH
`chocheol@gmail.com`
Postdoctoral supervisor